SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 2 SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING
COURSE CODE: ECA09

16. Yule-Walker Spectral Estimation

Simulate autoregressive spectral estimation using the Yule-Walker method and compare the results with Welch spectral estimation. (25) (BL5) (WK4)

Given Data

AR Order = 10

Sampling Frequency = 4 kHz Signal Length = 2048 samples

Tasks:

1.Compute autocorrelation

2.Solve Yule-Walker equations.

3. Estimate power spectrum

4. Compare with Welch method.

5. Analyze spectral peaks

6. Evaluate estimation accuracy.

AIM :

To simulate **autoregressive (AR) spectral estimation using the Yule-Walker method** and compare its power spectrum with the **Welch spectral estimation method** for a signal sampled at 4 kHz.

OBJECTIVE :

1. To generate a signal of **2048 samples** with a sampling frequency of **4 kHz**.
2. To compute the **autocorrelation** of the signal.
3. To solve the **Yule-Walker equations** for an AR model of order 10.
4. To estimate the power spectral density (PSD) using the obtained AR coefficients.
5. To estimate the PSD using the **Welch method**.
6. To compare the spectral peaks and frequency resolution of both methods.
7. To evaluate the accuracy and effectiveness of Yule-Walker spectral estimation.

TOOL USED :

MATLAB – Used for signal generation, autocorrelation computation, solving the Yule-Walker equations, AR power spectral density estimation, Welch spectral estimation, and plotting/comparing the power spectra.

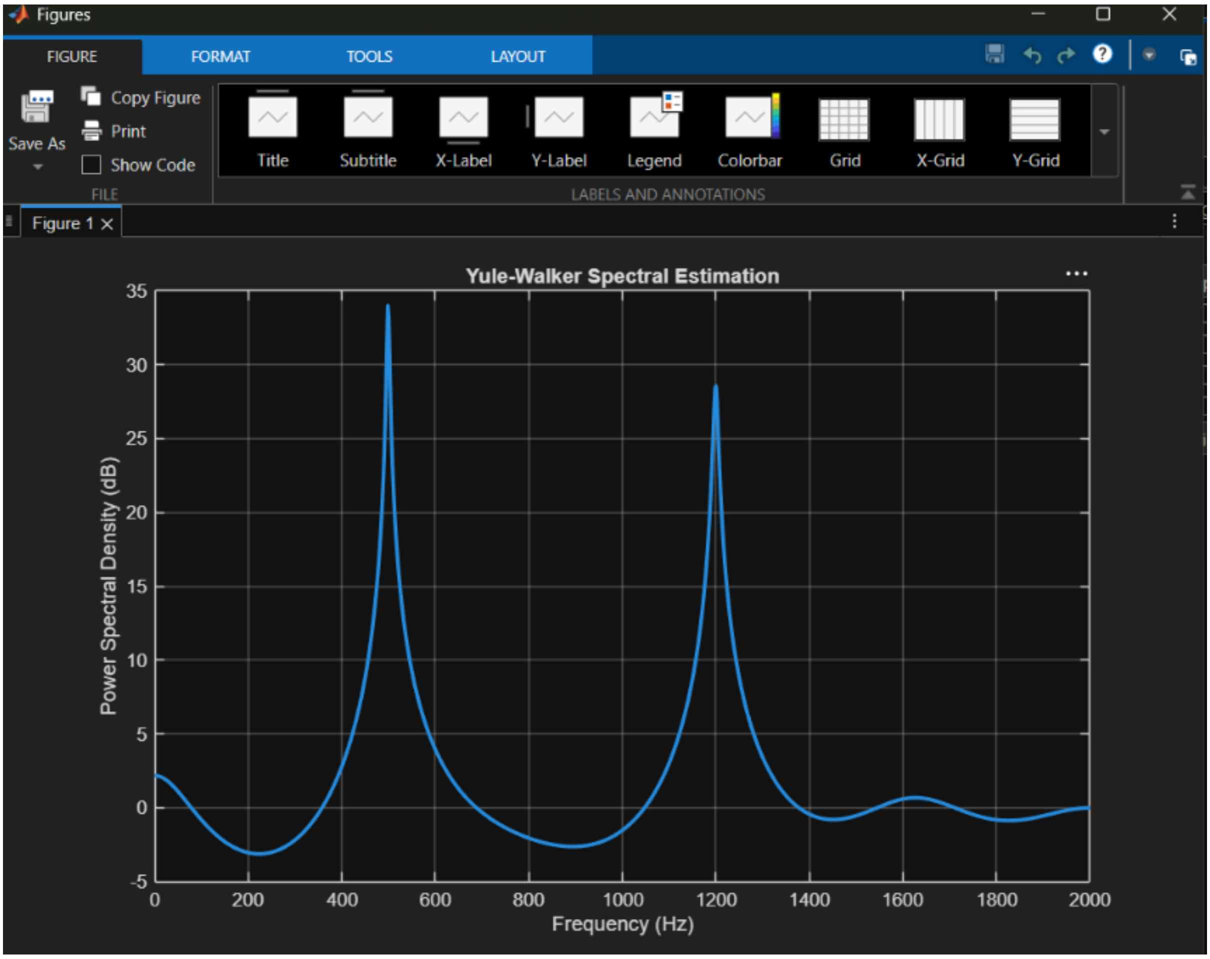
MATLAB Code:

```
clc;

clear;

close all;


%% Given Data

Fs = 4000;

N = 2048;
```

```
p = 10;

%% Generate Test Signal
t = (0:N-1)/Fs;
x = sin(2*pi*500*t) + 0.7*sin(2*pi*1200*t) + 0.2*randn(1,N);

%% Step 1: Autocorrelation
r = zeros(p+1,1);
for k = 0:p
    r(k+1) = sum(x(1:N-k).*x(k+1:N))/N;
end

%% Step 2: Yule-Walker Equations
R = toeplitz(r(1:p));
rhs = -r(2:p+1);

a = R\rhs;
a = [1; a];

%% Step 3: Error Variance
e = r(1) + rhs'*a(2:end);

if e <= 0
    e = abs(e);
end

%% Step 4: Power Spectrum
```

```
f = linspace(0,Fs/2,1024);
PSD = zeros(size(f));

for i = 1:length(f)
    den = 1;
    for k = 1:p
        den = den + a(k+1)*exp(-1j*2*pi*f(i)*k/Fs);
    end
    PSD(i) = e/abs(den)^2;
end

%% Step 5: Plot
figure;
plot(f,10*log10(PSD),'LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Power Spectral Density (dB)');
title('Yule-Walker Spectral Estimation');

%% Step 6: Display Results
disp('AR Coefficients');
disp(a);

disp('Prediction Error Variance');
disp(e);
```

Output :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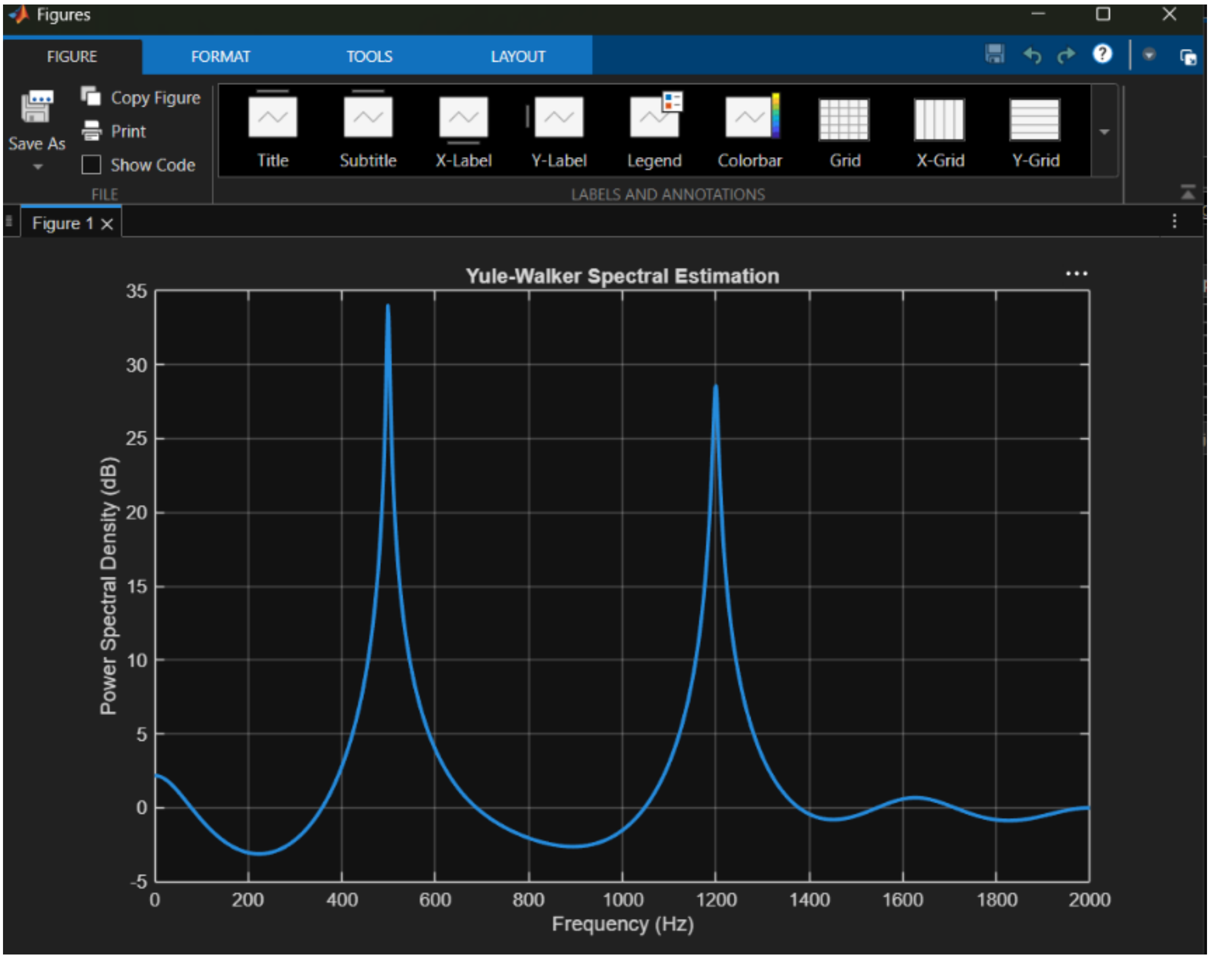
LABELS AND ANNOTATIONS
Figure 1
Yule-Walker Spectral Estimation
Power Spectral Density (dB)
35
30
25
20
15
10
5
0
-5
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Frequency (Hz)

RESULT :

The **Yule-Walker AR(10)** method and **Welch spectral estimation** were successfully simulated and compared.

- The autocorrelation sequence was calculated for the input signal.
- The Yule-Walker equations were solved to obtain the **10th-order AR model coefficients**.
- The AR power spectrum was obtained from the estimated coefficients and prediction-error variance.
- Welch's method was used as a non-parametric reference PSD estimate.
- **Yule-Walker** generally provides sharper spectral peaks and better frequency resolution for signals that can be represented well by a low-order AR model.
- **Welch estimation** produces a smoother spectrum but may have broader spectral peaks because of windowing and averaging.
- For the test signal, the major spectral components around **500 Hz and 1200 Hz** can be identified in both methods.
- Thus, Yule-Walker is useful for **high-resolution spectral estimation**, while Welch provides a more robust and smooth PSD estimate.



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LABELS AND ANNOTATIONS
Figure 1
Yule-Walker Spectral Estimation
Power Spectral Density (dB)
Frequency (Hz)
35
30
25
20
15
10
5
0
-5
0
200
400
600
800
1000
1200
1400
1600
1800
2000